

Problem Set 1: Verlet and Velocity-Verlet Integration

Consider a particle of mass m moving in a potential $V(\mathbf{r})$. Its equation of motion is

$$m\ddot{\mathbf{r}}(t) = \mathbf{f}(t), \quad \mathbf{f}(t) = -\nabla V(\mathbf{r}(t)). \quad (1)$$

For simplicity, suppress the particle index throughout this problem.

Part I: Deriving the Verlet Algorithm

- (a) Expand the exact position $\mathbf{r}(t + \Delta t)$ in a Taylor series about time t , retaining terms through third order:

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}(t)\Delta t + \frac{\mathbf{f}(t)}{2m}\Delta t^2 + \frac{1}{3!}\mathbf{r}^{(3)}(t)\Delta t^3 + O(\Delta t^4). \quad (2)$$

Similarly, expand $\mathbf{r}(t - \Delta t)$ about time t .

- (b) Derive

$$\mathbf{r}(t + \Delta t) = 2\mathbf{r}(t) - \mathbf{r}(t - \Delta t) + \frac{\mathbf{f}(t)}{m}\Delta t^2 + O(\Delta t^4).$$

(3)

This is the Verlet position-update equation.

Explain why the cancellation of the odd powers is important for the accuracy of the algorithm.

- (c) Use the same forward and backward Taylor expansions to derive a centered finite-difference expression for the velocity,

$$\mathbf{v}(t) \approx \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t - \Delta t)}{2\Delta t}. \quad (4)$$

Discuss one practical disadvantage of this formulation for molecular dynamics simulations.

Part II: Velocity-Verlet

The velocity-Verlet algorithm is defined by

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}(t)\Delta t + \frac{\mathbf{f}(t)}{2m}\Delta t^2, \quad (5)$$

$$\mathbf{v}(t + \Delta t) = \mathbf{v}(t) + \frac{\mathbf{f}(t + \Delta t) + \mathbf{f}(t)}{2m}\Delta t. \quad (6)$$

- (d) Starting from the velocity-Verlet equations, show that

$$\mathbf{r}(t + \Delta t) = 2\mathbf{r}(t) - \mathbf{r}(t - \Delta t) + \frac{\mathbf{f}(t)}{m}\Delta t^2.$$

(7)

Compare this result with the Verlet equation obtained in Part I.

What does this tell you about the position trajectory generated by velocity Verlet?

Part III: Time Reversibility

A molecular dynamics integrator is time reversible if reversing the velocity and propagating backward reproduces the previous trajectory.

Suppose that a velocity-Verlet step takes

$$(\mathbf{r}(t), \mathbf{v}(t)) \longrightarrow (\mathbf{r}(t + \Delta t), \mathbf{v}(t + \Delta t)). \quad (8)$$

The forward equations are

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}(t)\Delta t + \frac{\mathbf{f}(t)}{2m}\Delta t^2, \quad (9)$$

$$\mathbf{v}(t + \Delta t) = \mathbf{v}(t) + \frac{\mathbf{f}(t + \Delta t) + \mathbf{f}(t)}{2m}\Delta t. \quad (10)$$

(e) At time $t + \Delta t$, reverse the velocity:

$$\mathbf{v}(t + \Delta t) \rightarrow -\mathbf{v}(t + \Delta t). \quad (11)$$

Now apply the velocity-Verlet position update for one time step using the reversed velocity and the force $\mathbf{f}(t + \Delta t)$.

Show that the resulting position is

$$\boxed{\mathbf{r}'(t) = \mathbf{r}(t)}. \quad (12)$$

(f) Using the velocity-update equation, show that the velocity obtained after the reverse step is

$$\boxed{\mathbf{v}'(t) = -\mathbf{v}(t)}. \quad (13)$$

Problem Set 2: Enhanced Sampling and Free-Energy Calculations

Part I: Free-Energy Perturbation and the Overlap Problem

Consider two thermodynamic states A and B , with potential energies $U_A(\mathbf{x})$ and $U_B(\mathbf{x})$. Define

$$\Delta U(\mathbf{x}) = U_B(\mathbf{x}) - U_A(\mathbf{x}). \quad (14)$$

For lecture, we know

$$\Delta F_{BA} = -k_B T \ln \left\langle e^{-\beta \Delta U} \right\rangle_A. \quad (15)$$

- (a) Suppose that ΔU is approximately Gaussian in ensemble A ,

$$\Delta U \sim \mathcal{N}(\mu, \sigma^2), \quad (16)$$

so that

$$p_A(\Delta U) \propto \exp \left[-\frac{(\Delta U - \mu)^2}{2\sigma^2} \right]. \quad (17)$$

Determine the value of ΔU at which the integral for ΔF_{BA} is maximized. Show that

$$\Delta U_{\text{dom}} = \mu - \beta \sigma^2. \quad (18)$$

In other words, identify the configurations that make the largest contribution to the FEP exponential average.

- (b) The most probable value of ΔU in the original ensemble A is μ . Using your result from part (b), determine how many standard deviations the dominant value ΔU_{dom} lies below the mean.

Show that

$$\frac{\mu - \Delta U_{\text{dom}}}{\sigma} = \beta \sigma. \quad (19)$$

Explain why a large value of $\beta \sigma$ creates a sampling problem for FEP.

In particular, discuss why the configurations that dominate the exponential average can become extremely rare in the ensemble from which the simulation is actually sampled.

Part II: MBAR as Reweighting from a Mixture Distribution

Define the pooled mixture distribution

$$p_{\text{mix}}(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^M N_i p_i(\mathbf{x}). \quad (20)$$

Since

$$p_i(\mathbf{x}) = e^{f_i - u_i(\mathbf{x})}, \quad (21)$$

we have

$$p_{\text{mix}}(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^M N_i e^{f_i - u_i(\mathbf{x})}. \quad (22)$$

Derive an expression for

$$\langle A \rangle_0 = \int d\mathbf{x} p_0(\mathbf{x}) A(\mathbf{x}). \quad (23)$$

using $p_{\text{mix}}(\mathbf{x})$.

Part III: From MBAR to an Unbiased Free-Energy Profile

Suppose the quantity of interest is a collective variable

$$\xi = \xi(\mathbf{x}). \quad (24)$$

The unbiased probability distribution is

$$P_0(\xi) = \langle \delta[\xi - \xi(\mathbf{x})] \rangle_0. \quad (25)$$

Using the MBAR weights, derive an expression for $P_0(\xi)$.

Problem Set 3: Noise-Contrastive Estimation and Energy-Based Learning

In this problem set, we consider an energy-based model

$$p_{\theta}(\mathbf{x}) = \frac{e^{-\beta U_{\theta}(\mathbf{x})}}{Z_{\theta}}, \quad (26)$$

where

$$Z_{\theta} = \int d\mathbf{x} e^{-\beta U_{\theta}(\mathbf{x})} \quad (27)$$

is the configurational partition function.

We will compare maximum-likelihood estimation with noise-contrastive estimation (NCE).

The central question is:

Can we learn the same equilibrium distribution as maximum likelihood without explicitly evaluating or sampling from the model partition function?

 (28)

Part I: Constructing the NCE Classification Problem

Suppose that, in addition to data samples drawn from $p_{\text{data}}(\mathbf{x})$, we can efficiently generate samples from a known noise distribution

$$p_n(\mathbf{x}). \quad (29)$$

For every data sample, suppose that k noise samples are generated.

Let

$$y = 1 \quad (30)$$

denote a data sample and

$$y = 0 \quad (31)$$

denote a noise sample.

(a) Show that the class priors are

$$P(y = 1) = \frac{1}{1 + k}, \quad P(y = 0) = \frac{k}{1 + k}. \quad (32)$$

(b) Use Bayes' theorem to derive the true posterior probability

$$P(y = 1|\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + kp_n(\mathbf{x})}.$$

 (33)

(c) The NCE model predicts

$$q_{\theta}(\mathbf{x}) = P_{\theta}(y = 1|\mathbf{x}) = \frac{p_{\theta}(\mathbf{x})}{p_{\theta}(\mathbf{x}) + kp_n(\mathbf{x})}. \quad (34)$$

Show that the corresponding probability of the noise class is

$$1 - q_{\theta}(\mathbf{x}) = \frac{kp_n(\mathbf{x})}{p_{\theta}(\mathbf{x}) + kp_n(\mathbf{x})}. \quad (35)$$

Part II: Why Does NCE Recover the Data Distribution?

The population NCE loss is

$$\begin{aligned}\mathcal{L}_{\text{NCE}} = & - \int d\mathbf{x} p_{\text{data}}(\mathbf{x}) \log q_{\theta}(\mathbf{x}) \\ & - k \int d\mathbf{x} p_n(\mathbf{x}) \log [1 - q_{\theta}(\mathbf{x})].\end{aligned}\quad (36)$$

Rewrite the NCE loss as

$$\mathcal{L}_{\text{NCE}} = \int d\mathbf{x} \ell_{\mathbf{x}}(q), \quad (37)$$

where

$$\ell_{\mathbf{x}}(q) = -p_{\text{data}}(\mathbf{x}) \log q - k p_n(\mathbf{x}) \log(1 - q). \quad (38)$$

Treat q as an independent variable and minimize $\ell_{\mathbf{x}}(q)$ at fixed $\mathbf{x}$.

Show that the minimum occurs at

$$q_{\theta}(\mathbf{x}) = p_{\text{data}}(\mathbf{x}) \quad (39)$$

Part III: NCE in Terms of Energy Functions and Free Energy

Suppose that the noise distribution is itself an equilibrium distribution,

$$p_n(\mathbf{x}) = \frac{e^{-\beta U_n(\mathbf{x})}}{Z_n}, \quad (40)$$

while the model distribution is

$$p_{\theta}(\mathbf{x}) = \frac{e^{-\beta U_{\theta}(\mathbf{x})}}{Z_{\theta}}. \quad (41)$$

Define the free-energy difference

$$\Delta F = F_{\theta} - F_n = -k_{\text{B}} T \ln \frac{Z_{\theta}}{Z_n}. \quad (42)$$

(a) Show that

$$\boxed{\ln \frac{p_{\theta}(\mathbf{x})}{p_n(\mathbf{x})} = -\beta [U_{\theta}(\mathbf{x}) - U_n(\mathbf{x}) - \Delta F]}. \quad (43)$$

(b) Show that the NCE classifier can therefore be written as

$$\boxed{P_{\theta}(y = 1|\mathbf{x}) = \frac{1}{1 + k \exp \{\beta [U_{\theta}(\mathbf{x}) - U_n(\mathbf{x}) - \Delta F]\}}}. \quad (44)$$

(c) Explain why ΔF acts as a global offset or intercept in the classifier.

In particular, show from the above equation that increasing ΔF increases

$$P_{\theta}(y = 1|\mathbf{x}) \quad (45)$$

for every configuration $\mathbf{x}$.

- (d) The NCE loss contains contributions from both data and noise samples. Explain qualitatively what happens if

$$\Delta F \tag{46}$$

is chosen too large or too small.

Why can the two populations of samples determine the appropriate value of ΔF ?

- (e) Explain why ΔF can be treated as an independent fitting parameter during optimization even though its exact value is formally determined by the partition functions.

Part IV: Why Can NCE Treat the Free Energy as a Free Parameter?

Consider an energy-based model

$$p_{\theta}(\mathbf{x}) = \frac{e^{-\beta U_{\theta}(\mathbf{x})}}{Z_{\theta}}, \tag{47}$$

where

$$Z_{\theta} = \int d\mathbf{x} e^{-\beta U_{\theta}(\mathbf{x})}. \tag{48}$$

In NCE, we introduced a free-energy difference ΔF as an independent scalar parameter. Why can we do this in NCE, whereas we cannot simply introduce an arbitrary independent normalization parameter in maximum-likelihood estimation?

- (a) Write the negative log-likelihood for the energy-based model,

$$\mathcal{L}_{\text{ML}} = -\langle \log p_{\theta}(\mathbf{x}) \rangle_{\text{data}}, \tag{49}$$

and show that

$$\boxed{\mathcal{L}_{\text{ML}} = \beta \langle U_{\theta}(\mathbf{x}) \rangle_{\text{data}} + \log Z_{\theta}.} \tag{50}$$

Explain why Z_{θ} is not an independent parameter once $U_{\theta}(\mathbf{x})$ has been specified.

- (b) Suppose one nevertheless tries to introduce an independent scalar c and write

$$\tilde{p}_{\theta,c}(\mathbf{x}) = e^{-\beta U_{\theta}(\mathbf{x}) + c}. \tag{51}$$

Show that the corresponding negative log-likelihood would be

$$\mathcal{L}(\theta, c) = \beta \langle U_{\theta}(\mathbf{x}) \rangle_{\text{data}} - c. \tag{52}$$

What happens if c is treated as a completely free parameter?

Explain why the likelihood problem is ill-posed.

Problem Set 4: Affine Coupling Layers in Normalizing Flows

A normalizing flow seeks an invertible transformation

$$\mathbf{y} = f_{\theta}(\mathbf{x}) \quad (53)$$

between molecular configurations and a latent representation.

A major challenge is that the transformation must satisfy two requirements:

1. it must be invertible;
2. its Jacobian determinant must be computationally tractable.

A general neural network does not necessarily satisfy either requirement. An affine coupling layer is one way to construct a nonlinear transformation that satisfies both.

Suppose the input coordinates are partitioned into two subsets,

$$\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2). \quad (54)$$

Consider the following transformation:

$$\mathbf{y}_1 = \mathbf{x}_1, \quad (55)$$

$$\mathbf{y}_2 = \mathbf{x}_2 \odot \exp[\mathbf{s}(\mathbf{x}_1)] + \mathbf{t}(\mathbf{x}_1), \quad (56)$$

where $\odot$ denotes element-wise multiplication, and $\mathbf{s}(\mathbf{x}_1)$ and $\mathbf{t}(\mathbf{x}_1)$ are arbitrary differentiable functions, which may be represented by neural networks.

Part I: Deriving the Inverse Transformation

Derive an explicit expression for $\mathbf{x}_1$ and $\mathbf{x}_2$ in terms of $\mathbf{y}_1$ and $\mathbf{y}_2$.

Explain why the functions $\mathbf{s}(\mathbf{x}_1)$ and $\mathbf{t}(\mathbf{x}_1)$ do not themselves need to be invertible.

Part II: Computing the Jacobian

Define the Jacobian matrix

$$J = \frac{\partial \mathbf{y}}{\partial \mathbf{x}^T}. \quad (57)$$

Show that

$$\det J = \prod_j e^{s_j(\mathbf{x}_1)} = \exp \left[\sum_j s_j(\mathbf{x}_1) \right]. \quad (58)$$

Hence,

$$\log |\det J| = \sum_j s_j(\mathbf{x}_1). \quad (59)$$

Why is this much easier to evaluate than the determinant of a completely general high-dimensional Jacobian?

Part III: Why Are Multiple Coupling Layers Needed?

In a single coupling layer,

$$\mathbf{y}_1 = \mathbf{x}_1, \tag{60}$$

so the coordinates in $\mathbf{x}_1$ are not directly transformed.

Suppose a second coupling layer reverses the roles of the two subsets:

$$\mathbf{z}_2 = \mathbf{y}_2, \tag{61}$$

$$\mathbf{z}_1 = \mathbf{y}_1 \odot \exp [\tilde{\mathbf{s}}(\mathbf{y}_2)] + \tilde{\mathbf{t}}(\mathbf{y}_2). \tag{62}$$

Explain why stacking these two layers allows both subsets of coordinates to be transformed.

What advantage does changing the partition or permuting the coordinates between successive coupling layers provide?

Problem Set 5: Denoising Score Matching for Diffusion Models

Consider a diffusion process that starts from molecular configurations $\mathbf{x}_0$ sampled from the data distribution

$$\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x}_0). \quad (63)$$

At diffusion time t , the forward process defines a conditional distribution

$$p(\mathbf{x}_t | \mathbf{x}_0), \quad (64)$$

and the corresponding marginal distribution is

$$p_t(\mathbf{x}_t) = \int d\mathbf{x}_0 p_{\text{data}}(\mathbf{x}_0) p(\mathbf{x}_t | \mathbf{x}_0). \quad (65)$$

The goal of a diffusion model is to learn the score of the noisy marginal distribution,

$$\mathbf{s}_t(\mathbf{x}_t) = \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t). \quad (66)$$

Part I: Relating the Conditional and Marginal Scores

Define the conditional score

$$\mathbf{s}_c(\mathbf{x}_t, \mathbf{x}_0, t) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{x}_0). \quad (67)$$

(a) Starting from

$$p_t(\mathbf{x}_t) = \int d\mathbf{x}_0 p_{\text{data}}(\mathbf{x}_0) p(\mathbf{x}_t | \mathbf{x}_0), \quad (68)$$

differentiate with respect to $\mathbf{x}_t$ and show that

$$\nabla_{\mathbf{x}_t} p_t(\mathbf{x}_t) = \int d\mathbf{x}_0 p_{\text{data}}(\mathbf{x}_0) \nabla_{\mathbf{x}_t} p(\mathbf{x}_t | \mathbf{x}_0). \quad (69)$$

(b) Show that

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) = \mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t} [\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{x}_0)]. \quad (70)$$

Explain the physical or statistical meaning of this identity.

Part III: Denoising Score Matching

Because the conditional distribution $p(\mathbf{x}_t | \mathbf{x}_0)$ is chosen by the modeler, its score can often be evaluated analytically.

Define the denoising score-matching objective

$$\mathcal{L}_{\text{DSM}}(\theta) = \mathbb{E}_{\mathbf{x}_0, \mathbf{x}_t} [\|\mathbf{s}_\theta(\mathbf{x}_t, t) - \mathbf{s}_c(\mathbf{x}_t, \mathbf{x}_0, t)\|^2]. \quad (71)$$

(c) Expand the square and show that

$$\mathcal{L}_{\text{DSM}} = \mathbb{E} [\|\mathbf{s}_\theta\|^2] - 2\mathbb{E} [\mathbf{s}_\theta \cdot \mathbf{s}_c] + \mathbb{E} [\|\mathbf{s}_c\|^2]. \quad (72)$$

(d) Consider the cross term

$$\mathbb{E} [\mathbf{s}_\theta(\mathbf{x}_t, t) \cdot \mathbf{s}_c(\mathbf{x}_t, \mathbf{x}_0, t)]. \quad (73)$$

Show that

$$\mathbb{E} [\mathbf{s}_\theta \cdot \mathbf{s}_c] = \mathbb{E} [\mathbf{s}_\theta \cdot \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)]. \quad (74)$$

(e) Compare this with the ideal score-matching objective

$$\mathcal{L}_{\text{SM}} = \mathbb{E}_{\mathbf{x}_t} \left[\|\mathbf{s}_\theta(\mathbf{x}_t, t) - \mathbf{s}_c(\mathbf{x}_t, t)\|^2 \right]. \quad (75)$$

Show that

$$\boxed{\mathcal{L}_{\text{DSM}} = \mathcal{L}_{\text{SM}} + C(t)}, \quad (76)$$

where $C(t)$ is independent of θ .